



HADRONS

ELECTRICALS

Reliable Connections. Complete Safety.

WIRING HARNESSSES & BATTERY CABLES

BUILT TO CONNECT. BUILT TO PERFORM.



SAFE & RELIABLE



HIGH CONDUCTIVITY



FLAME RETARDANT



DURABLE & FLEXIBLE



OEM GRADE QUALITY

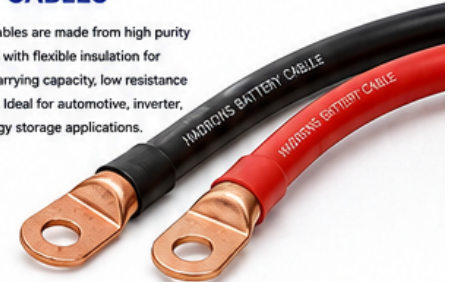
WIRING HARNESSSES

Hadrons Wiring Harnesses are precision engineered for home appliances to ensure reliable performance, easy installation and long service life. Manufactured with premium wires, terminals and connectors, our harnesses deliver safe power and signal transmission in every condition.



BATTERY CABLES

Hadrons Battery Cables are made from high purity copper conductors with flexible insulation for excellent current carrying capacity, low resistance and high durability. Ideal for automotive, inverter, industrial and energy storage applications.



TYPES OF WIRING HARNESSSES

WASHING MACHINE HARNESS



Complete harness for power supply, motor, drain pump, valve & control connections.

AIR CONDITIONER IDU HARNESS



Reliable connections for display, swing motor, sensors, fan motor & control PCB.

AIR CONDITIONER ODU HARNESS



Durable harness for compressor, fan motor, 4-way valve, sensors & PCB connections.

APPLIANCE & CONTROL HARNESS



For control panels, PCB, relays, switches & various electrical sub-assemblies.

SENSOR & SIGNAL HARNESS



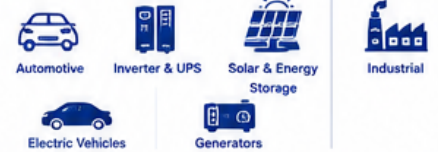
For temperature, pressure, NTC, thermistor & other sensor connections.

- High grade copper conductors
- Premium insulation (PVC / XLPE)
- Secure crimping & soldering
- Tidy & durable assembly
- 100% electrical testing
- RoHS compliant & Eco friendly

BATTERY CABLES - FEATURES

- ✓ High purity electrolytic copper for maximum conductivity
- ✓ Excellent flexibility for easy installation
- ✓ High current carrying capacity & low voltage drop
- ✓ Oil, abrasion, heat & weather resistant
- ✓ Available with ring terminal / eyelet / lug
- ✓ Custom lengths, colors & sizes available
- ✓ 100% tested for quality & performance

APPLICATIONS



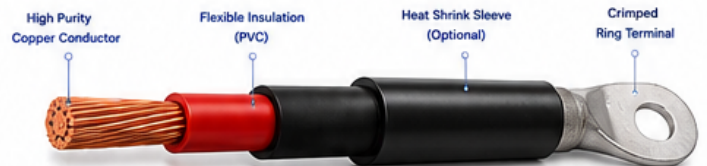
WIRING HARNESS CONSTRUCTION



TERMINAL & CONNECTOR OPTIONS



BATTERY CABLES CONSTRUCTION



TERMINAL OPTIONS



INSULATION COLOURS



WIRING HARNESSSES - TECHNICAL SPECIFICATIONS

Parameter	Specification
Conductor	Electrolytic Copper, Class 5 / Class 6
Insulation	PVC / XLPE (Flame Retardant)
Voltage Rating	Up to 600V AC / DC
Temperature Range	-25°C to +105°C
Flame Retardancy	VW-1 / IEC 60332-1
Bend Radius	As per wire diameter
Standards	IS 694 / ISO 6722 / RoHS Compliant
Testing	Continuity, Hi-Pot, Insulation Resistance, Visual

BATTERY CABLES - TECHNICAL SPECIFICATIONS

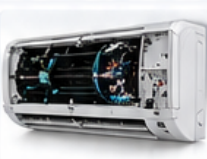
Conductor Cross Section (Sq. mm)	Construction (No. x Dia. mm)	Current Carrying Capacity (A)*	Insulation Thickness (mm)	Overall Diameter (mm)
16	126 x 0.40	110	1.40	8.0
25	196 x 0.40	150	1.60	9.5
35	276 x 0.40	185	1.70	10.5
50	396 x 0.40	230	1.80	12.0
70	360 x 0.50	290	2.00	13.5
95	475 x 0.50	360	2.20	15.5

*Current carrying capacity at 30°C ambient in open air

APPLICATIONS - HOME APPLIANCES



Washing Machine Complete Wiring Harness



Air Conditioner - IDU Wiring Harness



Air Conditioner - ODU Wiring Harness



Control & PCB Connections



Compressor & Fan Motor Wiring Harness

OEM CUSTOMIZATION

- ✓ Custom design as per drawing / sample
- ✓ Dedicated tooling & fixtures
- ✓ Low MOQ & on-time delivery
- ✓ 100% quality assurance
- ✓ Long term supply partnership



SAFE POWER. TOTAL PROTECTION.



PREMIUM QUALITY. BUILT TO LAST.



POWERING PERFORMANCE EVERY TIME.



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1. WIRING HARNESSES - CONSTRUCTION & APPLICATION

Wiring Harness Construction Table

Layer / Component	Typical Material / Option	Purpose
Conductor	Flexible electrolytic copper; fine stranded	Carries power or control signal
Primary Insulation	PVC / FR PVC / XLPE	Electrical insulation and temperature protection
Terminal	Brass / copper alloy; tin plated where required	Crimped electrical interface
Connector Housing	Nylon / engineering polymer	Terminal retention, polarization and mating
Splice / Joint	Crimp splice or soldered joint as drawing requires	Branching and circuit distribution
Protection	PVC sleeve, heat-shrink, braided sleeve, corrugated tube	Abrasion, heat and mechanical protection
Retention	Cable tie, clip, clamp, tape	Routing and assembly control
Identification	Wire colour, sleeve, label or marker	Traceability and circuit identification

Typical OEM Harness Applications

Harness Type	Typical Appliance Area	Typical Circuits / Interfaces
Washing Machine Harness	Main PCB, motor, pump, inlet valve, heater, door lock and sensors	Power, motor, drain pump, valve, heater, sensor and control connections
AC IDU Harness	Indoor PCB, display, blower motor, swing motor and thermistors	Low-voltage control, fan motor, sensor and display connections
AC ODU Harness	Outdoor PCB, compressor, fan motor, valves and sensors	Compressor / fan power, control, sensor and valve connections
Control Panel Harness	PCB, relay, switch, indicator and sub-assembly	Internal equipment interconnection
Sensor Harness	Temperature, pressure, level and position sensors	Low-current signal transmission

2. HARNESS TECHNICAL & CUSTOMIZATION OPTIONS

Parameter	Typical Catalogue Range / Option
Wire Size	0.35, 0.5, 0.75, 1.0, 1.5, 2.5 sq mm and customized sizes
Core / Branch Count	Single circuit to multi-branch assemblies as per customer drawing
Conductor	Flexible stranded copper
Insulation	PVC / FR PVC / XLPE as application requires
Voltage Class	Low-voltage signal circuits and appliance power circuits; finalized to OEM drawing
Termination	Open barrel terminals, ring terminals, spade terminals, pin terminals, receptacles and custom contacts
Connector Options	2-way onward; locking / polarized housings as per approved mating connector
Assembly Protection	Heat shrink, braided sleeve, corrugated conduit, PVC tubing and tapes
Testing	Continuity, polarity / circuit check, pull-force verification and visual inspection; HV / IR where specified
Customization	Wire length, colour, marking, connector, terminal, branch point and protection per OEM drawing / sample

3. BATTERY CABLES - CONSTRUCTION

Construction Element	Typical Option	Function
Conductor	Fine stranded electrolytic copper	High-current conduction with flexibility
Insulation	PVC / heat-resistant flexible compound	Electrical insulation and environmental protection
Terminal / Lug	Copper / tinned copper ring lug or OEM terminal	Secure bolted electrical connection
Crimp Joint	Controlled compression crimp	Low-resistance conductor-to-terminal joint
Heat Shrink	Optional, including colour identification	Joint insulation, strain relief and identification
Protective Sleeve	Optional braided / corrugated protection	Abrasion and routing protection

4. QUALITY CONTROL & OEM DEVELOPMENT

Quality Check	What It Verifies
Conductor / wire verification	Correct cross-section, colour, length and specification
Crimp height / visual check	Controlled terminal deformation and acceptable crimp geometry
Pull-force test	Mechanical retention of terminal-to-conductor joint
Continuity / circuit test	Correct electrical path and circuit mapping
Polarity / pin-position test	Correct terminal position and connector orientation
Insulation / HV test where specified	Electrical insulation integrity
Final visual inspection	Sleeve, routing, labels, locking, damage and workmanship

OEM Development Support

Development can be based on customer drawing, approved sample or mating-component details. Final BOM, wire specification, connector / terminal make, crimp parameters, testing criteria, packaging and traceability should be frozen through customer approval before mass production.

Applications: Washing machines | AC indoor units | AC outdoor units | appliance control panels | compressors & fan motors | inverter / UPS | automotive & EV auxiliary circuits | industrial DC equipment